# DS4420 Final Project: Bayesian Model

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## Bayesian Classification: Traffic Accident Severity

### Predicts severity (1-4) from environmental/situational features

### Feature Engineering

```r
library(brms)
```

```
## Warning: package 'brms' was built under R version 4.3.3

## Loading required package: Rcpp

## Loading 'brms' package (version 2.22.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').

##
## Attaching package: 'brms'

## The following object is masked from 'package:stats':
##
##     ar
```

```r
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
# load cleaned data
accidents <- as.data.frame(read_csv("accidents_clean.csv"))
```

```
## Rows: 10000 Columns: 14

## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (3): Weather_Condition, Sunrise_Sunset, State
## dbl  (9): Severity, Temperature(F), Humidity(%), Visibility(mi), Distance(mi...
## dttm (1): Start_Time
## date (1): Date
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# feature engineering
# bin hour into time of day
accidents$TimeOfDay <- cut(accidents$Hour,
                           breaks = c(-1, 5, 11, 16, 20, 23),
                           labels = c("Night", "Morning", "Afternoon", "Evening", "LateNight"))

# simplify weather into broad buckets
accidents$Weather_Bucket <- ifelse(grepl("Rain|Drizzle|Shower|Thunder|T-Storm", accidents$Weather_Condi
                            ifelse(grepl("Snow|Sleet|Hail|Ice|Wintry", accidents$Weather_Condition, igno
                            ifelse(grepl("Fog|Mist|Haze|Smoke", accidents$Weather_Condition, ignore.case
                            ifelse(grepl("Cloud|Overcast", accidents$Weather_Condition, ignore.case=T),
                            ifelse(grepl("Clear|Fair|Sunny",accidents$Weather_Condition, ignore.case=T)

# brms needs categorical variables as factors, not integers
accidents$IsNight <- as.factor(accidents$Sunrise_Sunset == "Night")
accidents$Severity <- as.factor(accidents$Severity)
accidents$Weather_Bucket <- as.factor(accidents$Weather_Bucket)
accidents$TimeOfDay <- as.factor(accidents$TimeOfDay)

# scale continuous features
accidents$Temp <- scale(accidents$`Temperature(F)`)
accidents$Humidity <- scale(accidents$`Humidity(%)`)
accidents$Visibility <- scale(accidents$`Visibility(mi)`)
accidents$Distance <- scale(accidents$`Distance(mi)`)

# remove other since it doesn't end up being included in test sample
accidents <- accidents[accidents$Weather_Bucket != "Other", ]
accidents$Weather_Bucket <- droplevels(accidents$Weather_Bucket)

# downsample to balance classes (severity 2 and 3 dominate)
set.seed(42)
balanced <- do.call(rbind, lapply(1:4, function(k) {
  s <- accidents[accidents$Severity == k, ]
  target <- case_when(
    # keep severity 1 & 4 as is, downsample 2 & 3 so it evens out dataset
    # for mdoeling purposes
    k == 1 ~ 81,
    k == 2 ~ 800,
    k == 3 ~ 800,
    k == 4 ~ 207
  )
  s[sample(nrow(s), min(nrow(s), target)), ]
}))

# train/test split (80/20)
train_idx <- sample(nrow(balanced), floor(0.8 * nrow(balanced)))
train <- balanced[train_idx, ]
test  <- balanced[-train_idx, ]

print(table(balanced$Severity))
```

```
##
##   1   2   3   4
##  81 800 800 207
```

**Fitting the model**

```r
# check default priors
default_prior(Severity ~ Temp + Humidity + Visibility + Distance + IsNight + Weather_Bucket + TimeOfDay
              data = train,
              family = categorical("logit"))
```

```
##                 prior     class                 coef group resp dpar nlpar lb
##                (flat)         b                                     mu2
##                (flat)         b             Distance                mu2
##                (flat)         b             Humidity                mu2
##                (flat)         b          IsNightTRUE                mu2
##                (flat)         b                 Temp                mu2
##                (flat)         b   TimeOfDayAfternoon                mu2
##                (flat)         b     TimeOfDayEvening                mu2
##                (flat)         b   TimeOfDayLateNight                mu2
##                (flat)         b     TimeOfDayMorning                mu2
##                (flat)         b           Visibility                mu2
##                (flat)         b Weather_BucketCloudy                mu2
##                (flat)         b    Weather_BucketFog                mu2
##                (flat)         b   Weather_BucketRain                mu2
##                (flat)         b   Weather_BucketSnow                mu2
##  student_t(3, 0, 2.5) Intercept                                     mu2
##                (flat)         b                                     mu3
##                (flat)         b             Distance                mu3
##                (flat)         b             Humidity                mu3
##                (flat)         b          IsNightTRUE                mu3
##                (flat)         b                 Temp                mu3
##                (flat)         b   TimeOfDayAfternoon                mu3
##                (flat)         b     TimeOfDayEvening                mu3
##                (flat)         b   TimeOfDayLateNight                mu3
##                (flat)         b     TimeOfDayMorning                mu3
##                (flat)         b           Visibility                mu3
##                (flat)         b Weather_BucketCloudy                mu3
##                (flat)         b    Weather_BucketFog                mu3
##                (flat)         b   Weather_BucketRain                mu3
##                (flat)         b   Weather_BucketSnow                mu3
##  student_t(3, 0, 2.5) Intercept                                     mu3
##                (flat)         b                                     mu4
##                (flat)         b             Distance                mu4
##                (flat)         b             Humidity                mu4
##                (flat)         b          IsNightTRUE                mu4
##                (flat)         b                 Temp                mu4
##                (flat)         b   TimeOfDayAfternoon                mu4
##                (flat)         b     TimeOfDayEvening                mu4
##                (flat)         b   TimeOfDayLateNight                mu4
##                (flat)         b     TimeOfDayMorning                mu4
##                (flat)         b           Visibility                mu4
##                (flat)         b Weather_BucketCloudy                mu4
##                (flat)         b    Weather_BucketFog                mu4
```

```
##                (flat)         b   Weather_BucketRain             mu4
##                (flat)         b   Weather_BucketSnow             mu4
##  student_t(3, 0, 2.5) Intercept                                   mu4
##  ub       source
##          default
##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
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##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##          default
##          default
##     (vectorized)
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##     (vectorized)
##     (vectorized)
##     (vectorized)
##          default
##          default
##     (vectorized)
##     (vectorized)
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##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##     (vectorized)
##          default
```

```r
# define manual priors
manual_prior <- c(
  brms::prior(student_t(3, 0, 5), class = "Intercept", dpar = "mu2"),
  brms::prior(student_t(3, 0, 5), class = "Intercept", dpar = "mu3"),
```

```r
  brms::prior(student_t(3, 0, 5), class = "Intercept", dpar = "mu4"),
  brms::prior(normal(0, 1), class = "b", dpar = "mu2"),
  brms::prior(normal(0, 1), class = "b", dpar = "mu3"),
  brms::prior(normal(0, 1), class = "b", dpar = "mu4")
)

# fit the model
severity_brm <- brm(
  Severity ~ Temp + Humidity + Visibility + Distance + IsNight + Weather_Bucket + TimeOfDay,
  family  = categorical("logit"),
  data    = train,
  chains  = 4,
  iter    = 2000,
  warmup  = 500,
  thin    = 2,
  cores   = getOption("mc.cores", 4),
  prior   = manual_prior
)
```

```
## Compiling Stan program...

## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.3.2)'
## using SDK: 'MacOSX26.2.sdk'
## clang -arch x86_64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Library/Frame
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/library/StanHead
## In file included from /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/library/RcppEige
## In file included from /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/library/RcppEige
## /Library/Frameworks/R.framework/Versions/4.3-x86_64/Resources/library/RcppEigen/include/Eigen/src/Co
##   679 | #include <cmath>
##       |          ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1

## Start sampling
```

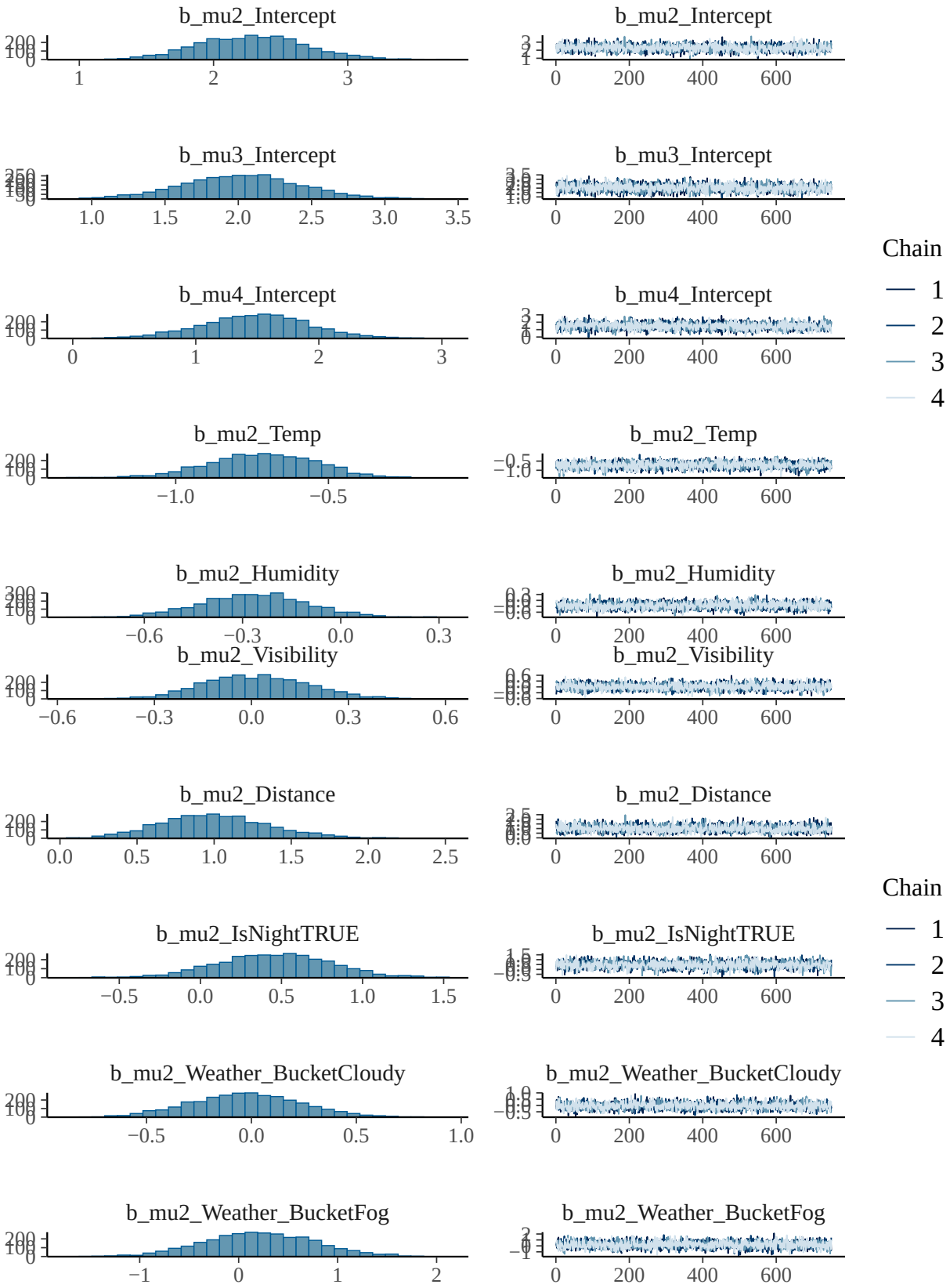
```r
# diagnostics
summary(severity_brm)
```

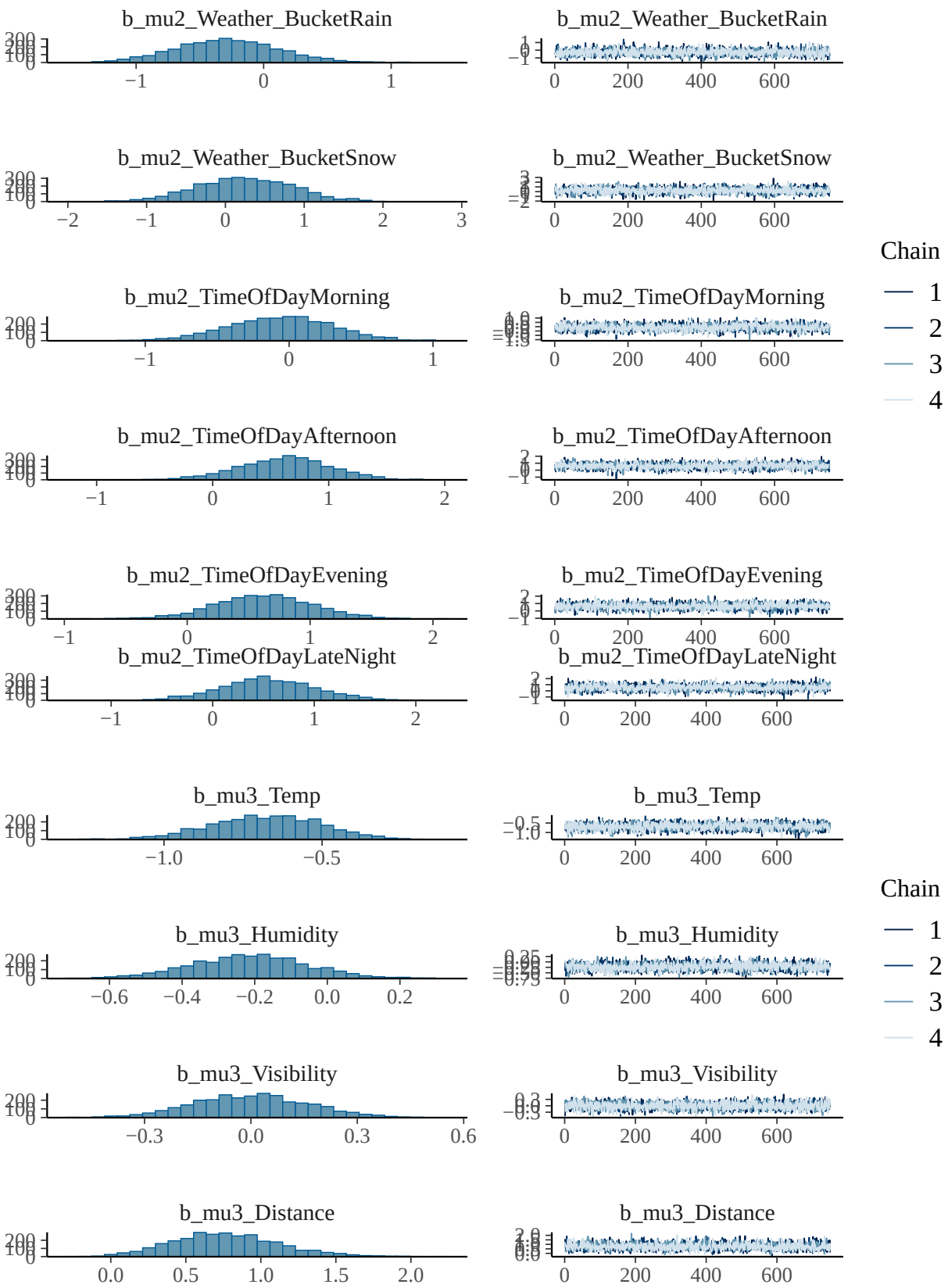
```
##  Family: categorical
##   Links: mu2 = logit; mu3 = logit; mu4 = logit
## Formula: Severity ~ Temp + Humidity + Visibility + Distance + IsNight + Weather_Bucket + TimeOfDay
##    Data: train (Number of observations: 1510)
##   Draws: 4 chains, each with iter = 2000; warmup = 500; thin = 2;
##          total post-warmup draws = 3000
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## mu2_Intercept                2.29      0.40     1.53     3.09 1.00     2591
## mu3_Intercept                2.02      0.40     1.24     2.83 1.00     2391
## mu4_Intercept                1.49      0.44     0.62     2.34 1.00     2508
## mu2_Temp                    -0.72      0.17    -1.07    -0.39 1.00     2561
## mu2_Humidity                -0.26      0.16    -0.58     0.05 1.00     2667
```

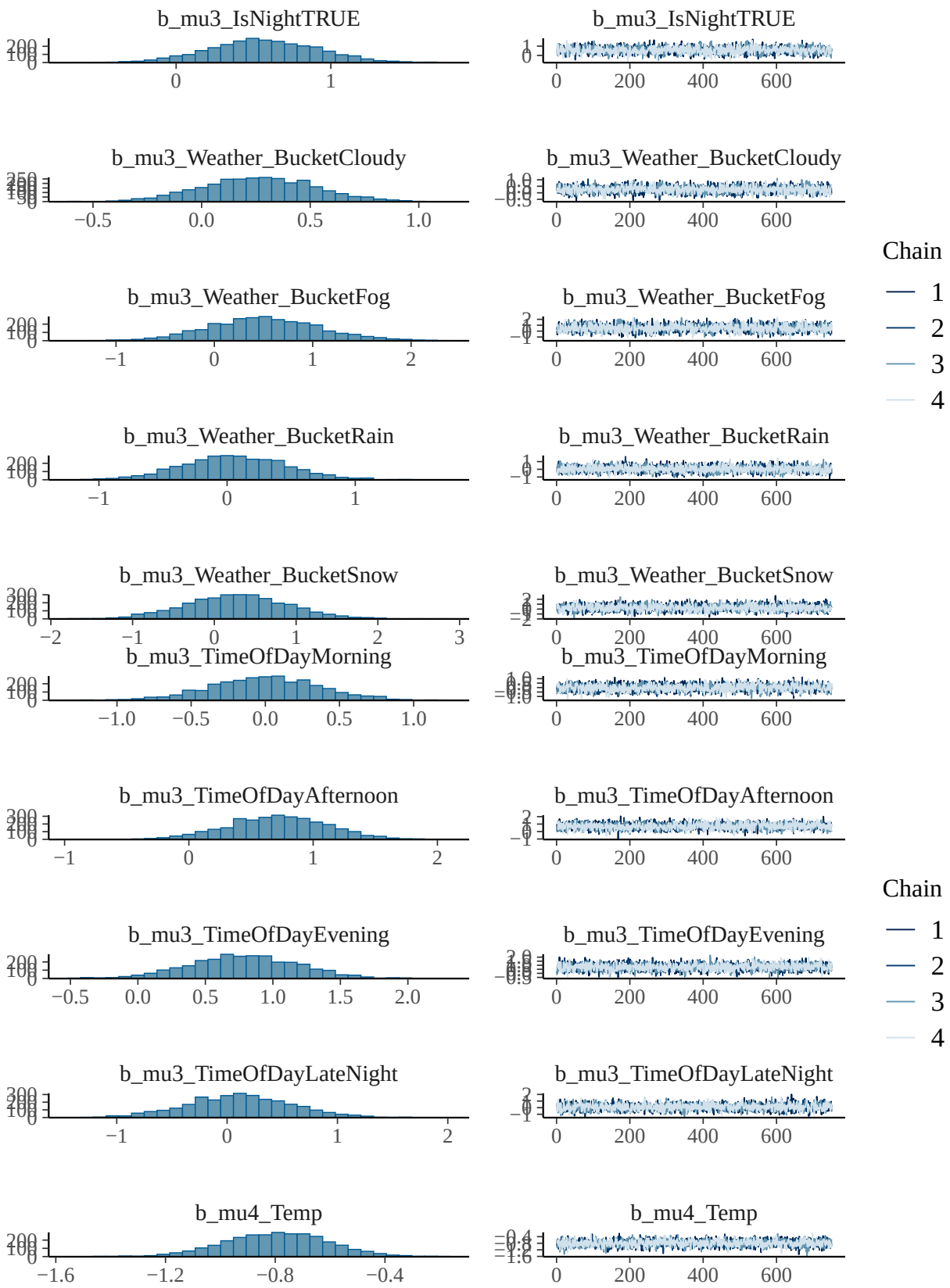
```
## mu2_Visibility               0.01      0.16    -0.29     0.33 1.00     2449
## mu2_Distance                 1.01      0.35     0.35     1.74 1.00     2540
## mu2_IsNightTRUE              0.46      0.35    -0.24     1.13 1.00     2573
## mu2_Weather_BucketCloudy    -0.03      0.26    -0.53     0.48 1.00     2549
## mu2_Weather_BucketFog        0.21      0.56    -0.85     1.32 1.00     2729
## mu2_Weather_BucketRain      -0.28      0.40    -1.06     0.52 1.00     2629
## mu2_Weather_BucketSnow       0.25      0.61    -0.93     1.44 1.00     2741
## mu2_TimeOfDayMorning        -0.06      0.36    -0.79     0.63 1.00     2609
## mu2_TimeOfDayAfternoon       0.62      0.39    -0.17     1.37 1.00     2612
## mu2_TimeOfDayEvening         0.63      0.40    -0.17     1.41 1.00     2548
## mu2_TimeOfDayLateNight       0.55      0.47    -0.38     1.48 1.00     2476
## mu3_Temp                    -0.68      0.17    -1.03    -0.35 1.00     2576
## mu3_Humidity                -0.21      0.16    -0.53     0.10 1.00     2714
## mu3_Visibility               0.00      0.16    -0.31     0.32 1.00     2663
## mu3_Distance                 0.77      0.35     0.12     1.52 1.00     2530
## mu3_IsNightTRUE              0.57      0.35    -0.10     1.25 1.00     2460
## mu3_Weather_BucketCloudy     0.26      0.26    -0.24     0.77 1.00     2518
## mu3_Weather_BucketFog        0.51      0.56    -0.57     1.66 1.00     2834
## mu3_Weather_BucketRain       0.06      0.40    -0.72     0.85 1.00     2703
## mu3_Weather_BucketSnow       0.28      0.62    -0.92     1.50 1.00     2810
## mu3_TimeOfDayMorning         0.00      0.36    -0.73     0.72 1.00     2550
## mu3_TimeOfDayAfternoon       0.71      0.40    -0.07     1.48 1.00     2582
## mu3_TimeOfDayEvening         0.79      0.40     0.02     1.57 1.00     2610
## mu3_TimeOfDayLateNight       0.11      0.47    -0.81     1.06 1.00     2692
## mu4_Temp                    -0.80      0.19    -1.17    -0.44 1.00     2618
## mu4_Humidity                -0.13      0.18    -0.47     0.23 1.00     2659
## mu4_Visibility              -0.09      0.19    -0.45     0.27 1.00     2653
## mu4_Distance                 1.17      0.35     0.52     1.91 1.00     2506
## mu4_IsNightTRUE              0.32      0.39    -0.47     1.08 1.00     2830
## mu4_Weather_BucketCloudy    -0.18      0.30    -0.77     0.40 1.00     2606
## mu4_Weather_BucketFog       -0.93      0.71    -2.39     0.45 1.00     2561
## mu4_Weather_BucketRain      -0.19      0.45    -1.07     0.70 1.00     2818
## mu4_Weather_BucketSnow      -0.15      0.66    -1.42     1.14 1.00     2808
## mu4_TimeOfDayMorning        -0.68      0.41    -1.46     0.12 1.00     2609
## mu4_TimeOfDayAfternoon      -0.19      0.44    -1.08     0.66 1.00     2572
## mu4_TimeOfDayEvening         0.43      0.43    -0.42     1.28 1.00     2663
## mu4_TimeOfDayLateNight       0.27      0.51    -0.70     1.29 1.00     2684
##                          Tail_ESS
## mu2_Intercept                2845
## mu3_Intercept                2687
## mu4_Intercept                2637
## mu2_Temp                     2615
## mu2_Humidity                 2625
## mu2_Visibility               2561
## mu2_Distance                 2834
## mu2_IsNightTRUE              2892
## mu2_Weather_BucketCloudy     2652
## mu2_Weather_BucketFog        1969
## mu2_Weather_BucketRain       2782
## mu2_Weather_BucketSnow       2727
## mu2_TimeOfDayMorning         2629
## mu2_TimeOfDayAfternoon       2762
## mu2_TimeOfDayEvening         2760
## mu2_TimeOfDayLateNight       2517
```

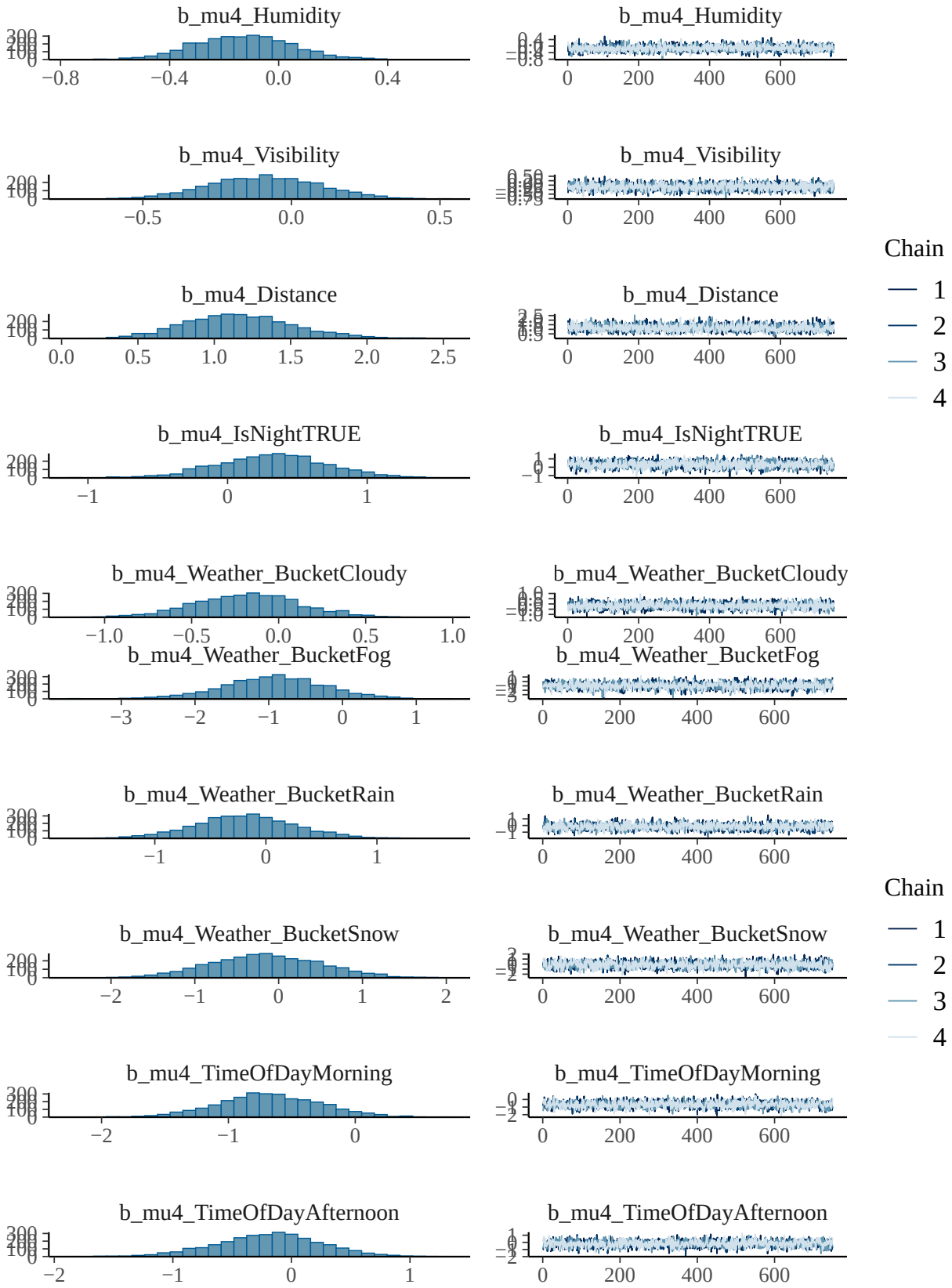
```
## mu3_Temp                     2782
## mu3_Humidity                 2760
## mu3_Visibility               2700
## mu3_Distance                 2872
## mu3_IsNightTRUE              2659
## mu3_Weather_BucketCloudy     2346
## mu3_Weather_BucketFog        2554
## mu3_Weather_BucketRain       2830
## mu3_Weather_BucketSnow       2688
## mu3_TimeOfDayMorning         2435
## mu3_TimeOfDayAfternoon       2723
## mu3_TimeOfDayEvening         2613
## mu3_TimeOfDayLateNight       2537
## mu4_Temp                     2692
## mu4_Humidity                 2736
## mu4_Visibility               3076
## mu4_Distance                 2797
## mu4_IsNightTRUE              2792
## mu4_Weather_BucketCloudy     2481
## mu4_Weather_BucketFog        2622
## mu4_Weather_BucketRain       2662
## mu4_Weather_BucketSnow       2533
## mu4_TimeOfDayMorning         2621
## mu4_TimeOfDayAfternoon       2640
## mu4_TimeOfDayEvening         2652
## mu4_TimeOfDayLateNight       2635
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

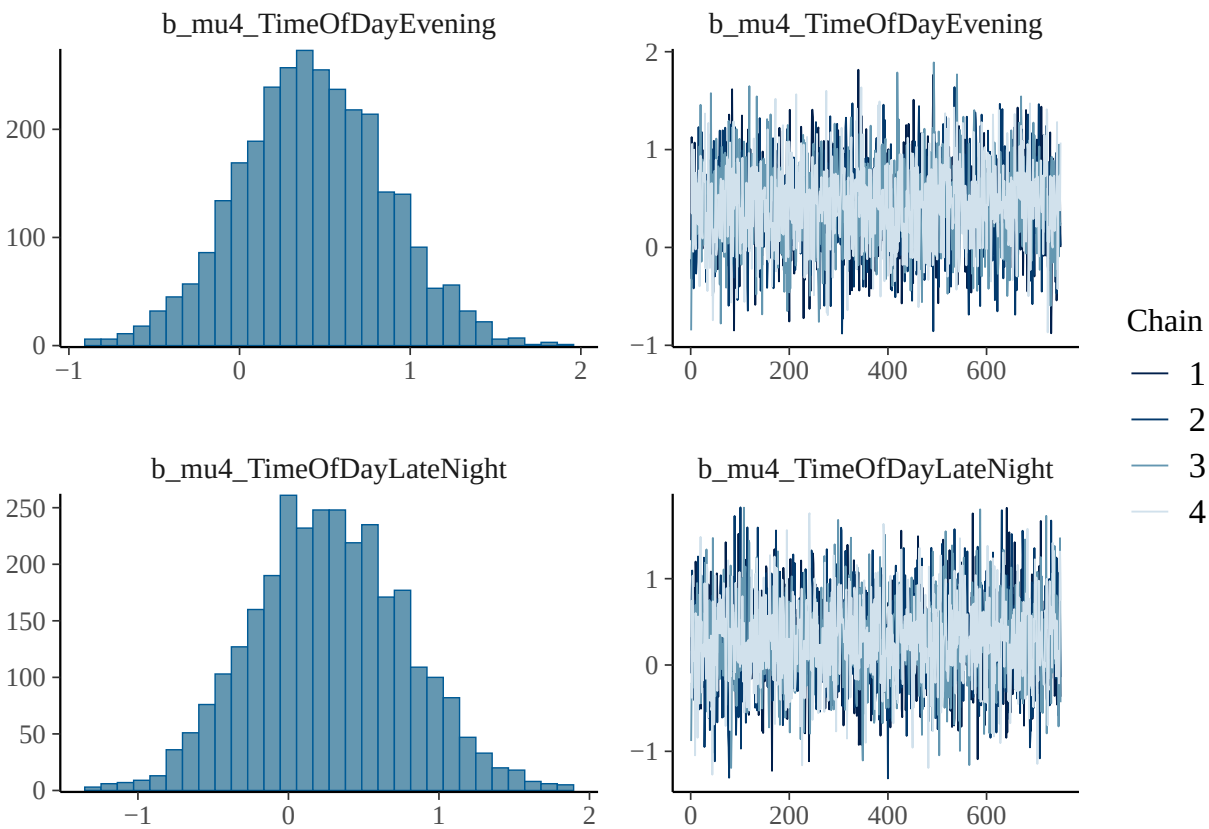
```r
plot(severity_brm)
```


b_mu2_Intercept
b_mu3_Intercept
b_mu4_Intercept
b_mu2_Temp
b_mu2_Humidity
b_mu2_Visibility
b_mu2_Distance
b_mu2_IsNightTRUE
b_mu2_Weather_BucketCloudy
b_mu2_Weather_BucketFog
Chain
1
2
3
4


b_mu2_Weather_BucketRain
b_mu2_Weather_BucketSnow
b_mu2_TimeOfDayMorning
b_mu2_TimeOfDayAfternoon
b_mu2_TimeOfDayEvening
b_mu2_TimeOfDayLateNight
b_mu3_Temp
b_mu3_Humidity
b_mu3_Visibility
b_mu3_Distance
Chain
1
2
3
4


b_mu3_IsNightTRUE
b_mu3_Weather_BucketCloudy
b_mu3_Weather_BucketFog
b_mu3_Weather_BucketRain
b_mu3_Weather_BucketSnow
b_mu3_TimeOfDayMorning
b_mu3_TimeOfDayAfternoon
b_mu3_TimeOfDayEvening
b_mu3_TimeOfDayLateNight
b_mu4_Temp
Chain
1
2
3
4


b_mu4_Humidity
b_mu4_Visibility
b_mu4_Distance
b_mu4_IsNightTRUE
b_mu4_Weather_BucketCloudy
b_mu4_Weather_BucketFog
b_mu4_Weather_BucketRain
b_mu4_Weather_BucketSnow
b_mu4_TimeOfDayMorning
b_mu4_TimeOfDayAfternoon
Chain
1
2
3
4




**Printing out metrics**

```r
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.3.3
## Loading required package: ggplot2
## Loading required package: lattice
## Registered S3 method overwritten by 'future':
##   method               from
##   all.equal.connection parallelly
```

```r
# posterior predictive for one accident
# each row is an MCMC sample, each column a test observation
# values are predicted severity class labels
post_preds <- posterior_predict(severity_brm, newdata=test)

# accuracy on test set
pred_classes <- apply(post_preds, 2, function(x) as.integer(names(which.max(table(x)))))
cat("Accuracy:", round(mean(pred_classes == as.integer(test$Severity)), 4), "\n")
```

```
## Accuracy: 0.4815
```

```r
# confusion matrix
print(table(Predicted=pred_classes, Actual=as.integer(test$Severity)))
```

```
##           Actual
## Predicted   1   2   3   4
```

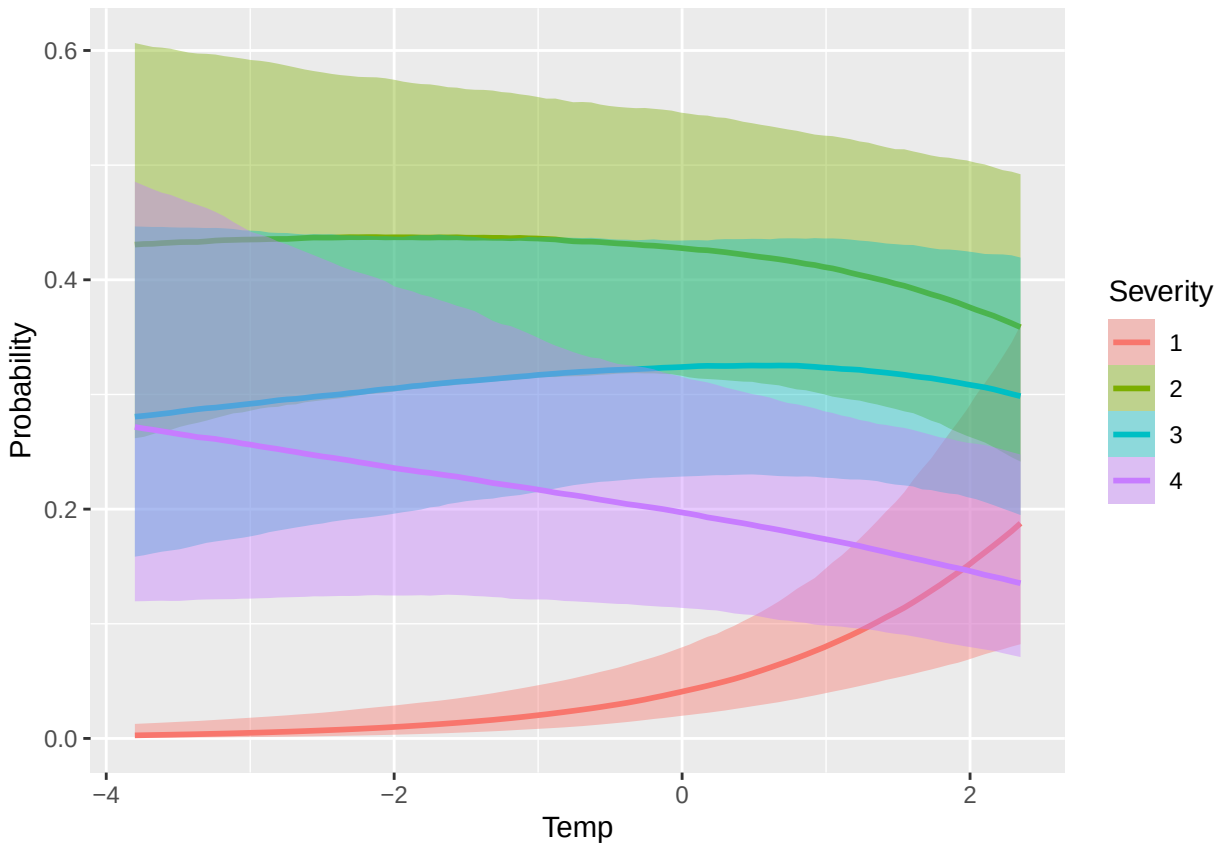
```
##         2   8  76  65  23
##         3   6  86 105   8
##         4   0   0   0   1
```

```r
# matrix for precision, accuracy, and f1 scores
cm <- confusionMatrix(
  factor(pred_classes, levels = 1:4),
  factor(as.integer(test$Severity), levels = 1:4)
)

# extract and print per-class metrics
metrics <- data.frame(
  Class     = paste("Severity", 1:4),
  Precision = round(cm$byClass[, "Precision"], 3),
  Recall    = round(cm$byClass[, "Recall"], 3),
  F1        = round(cm$byClass[, "F1"], 3)
)

print(metrics)
```

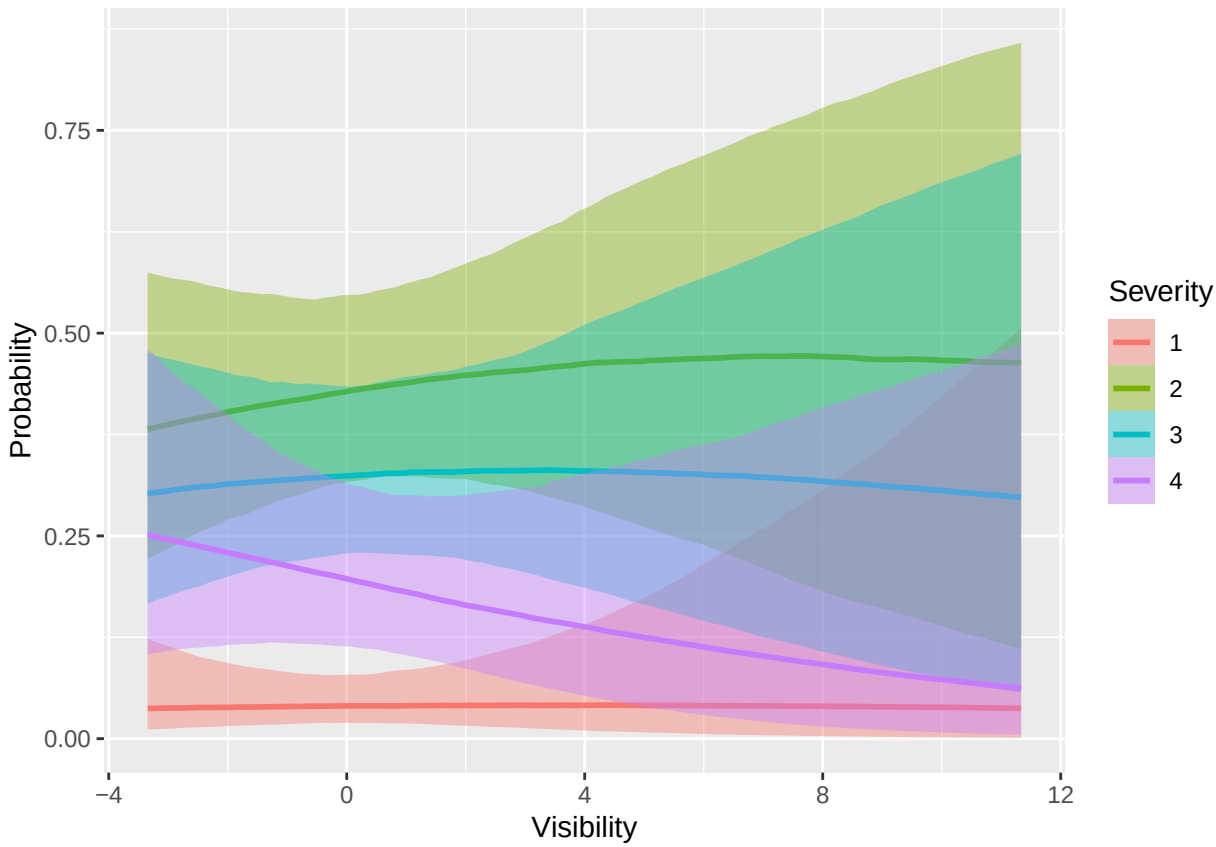
```
##               Class Precision Recall    F1
## Class: 1 Severity 1        NA  0.000    NA
## Class: 2 Severity 2     0.442  0.469 0.455
## Class: 3 Severity 3     0.512  0.618 0.560
## Class: 4 Severity 4     1.000  0.031 0.061
```

### Plotting

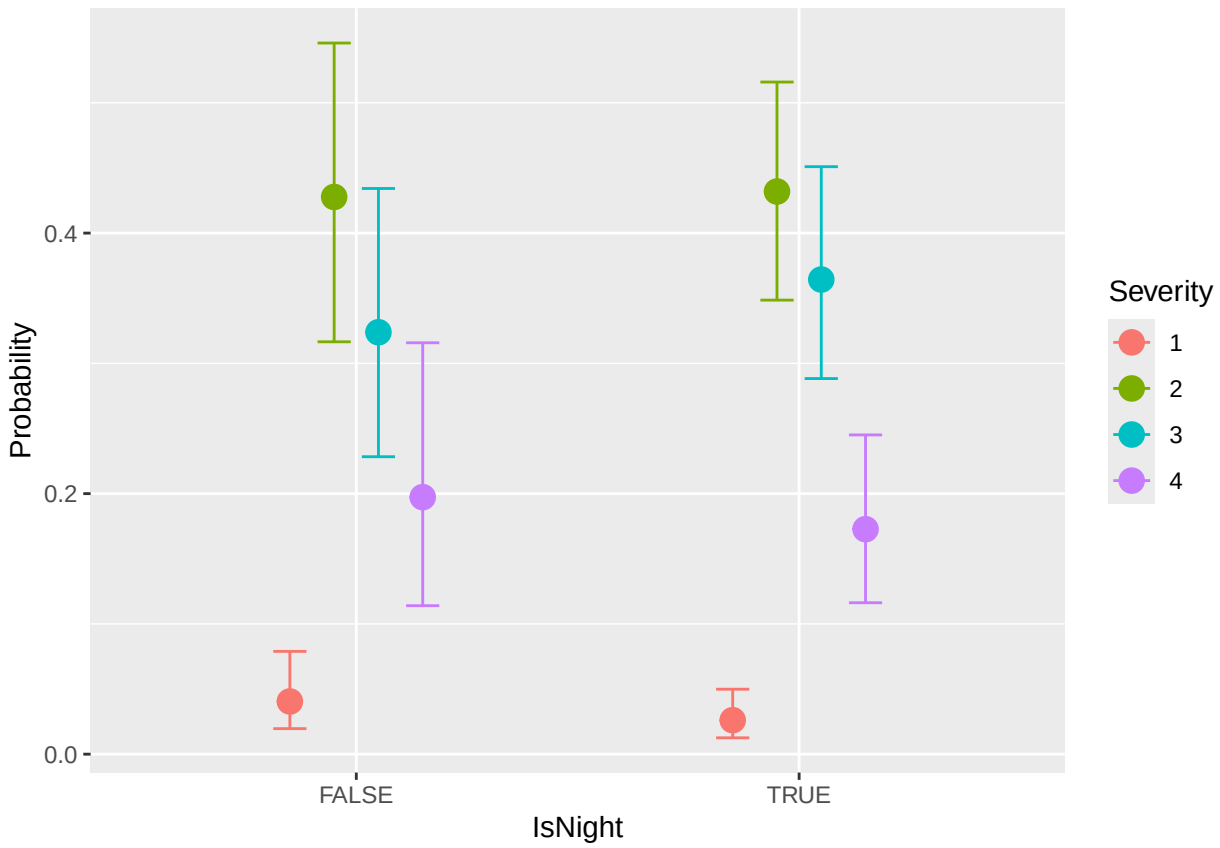
```r
# conditional effects - plot all to see the effects
plot(conditional_effects(severity_brm, "Temp", categorical=TRUE))
```



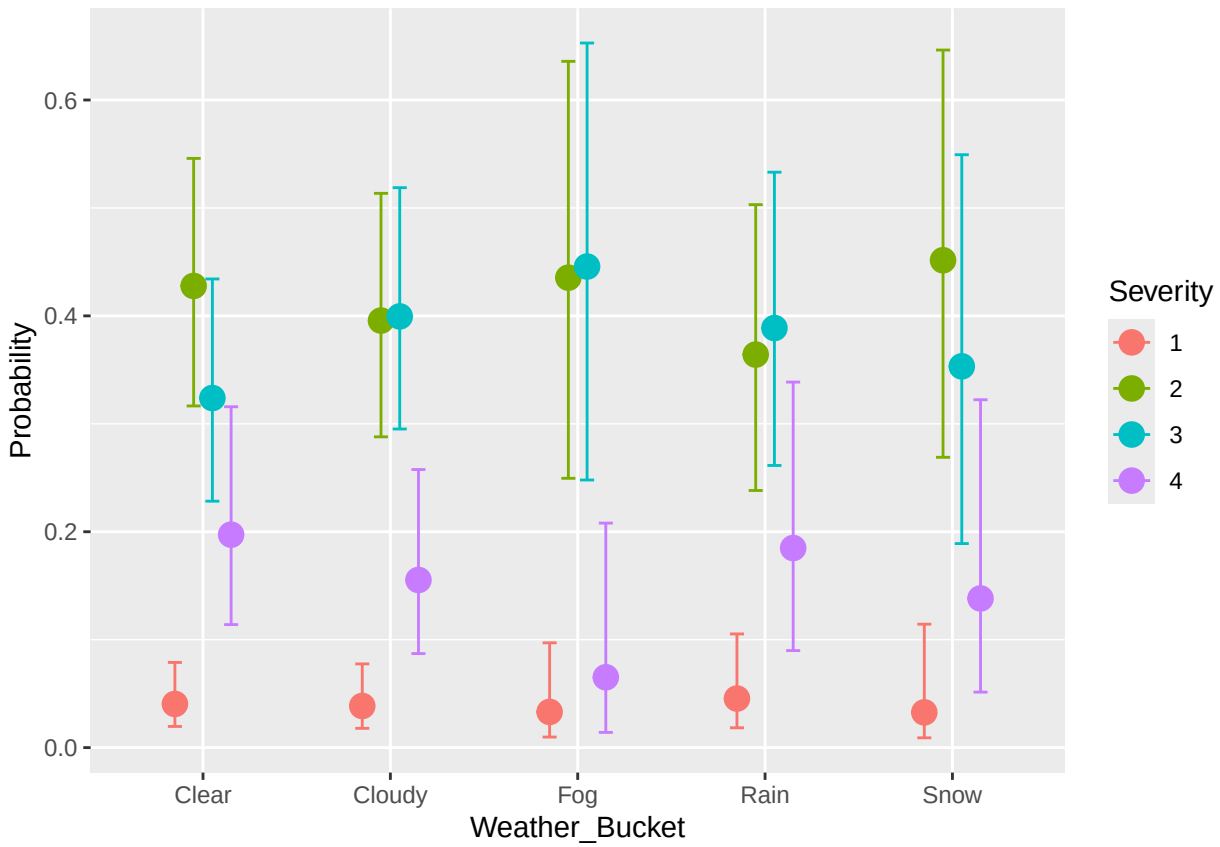

```
plot(conditional_effects(severity_brm, "Visibility", categorical=TRUE))
```



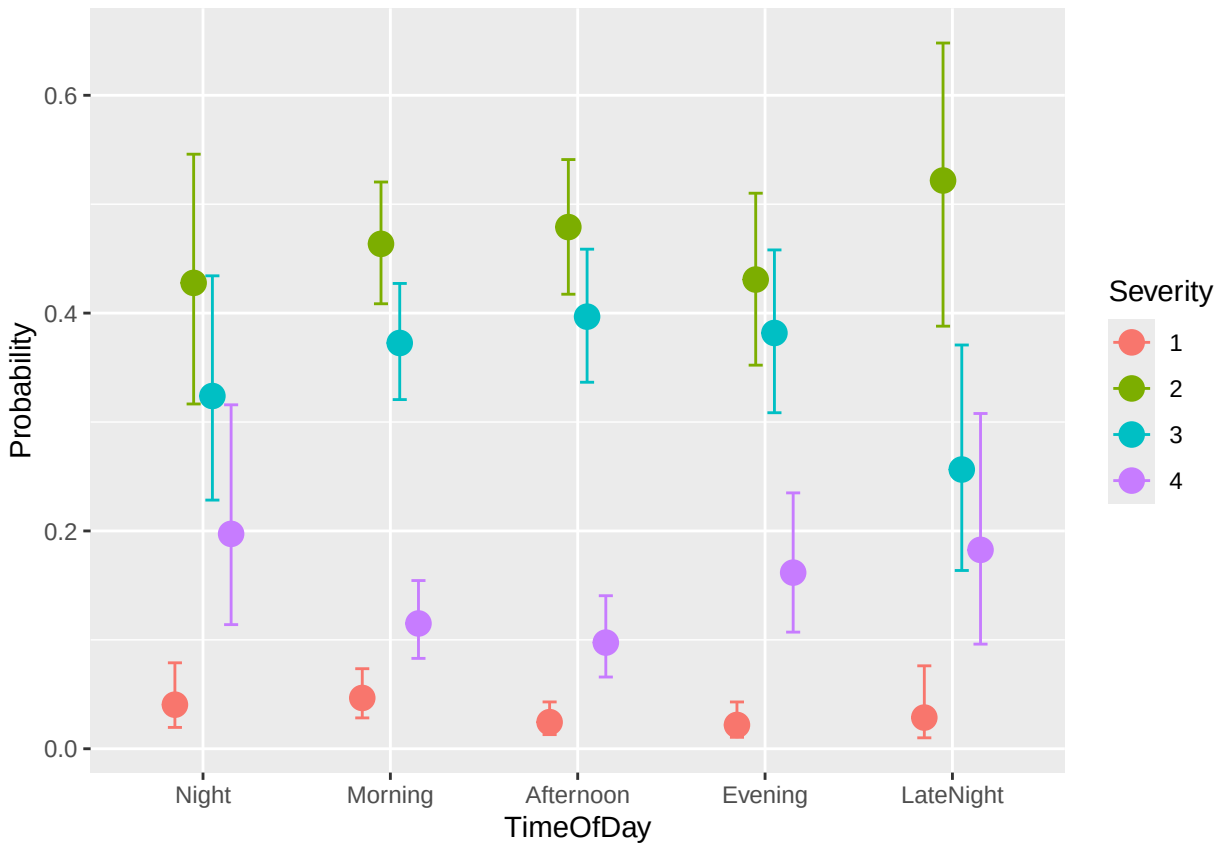

```
plot(conditional_effects(severity_brm, "IsNight", categorical=TRUE))
```



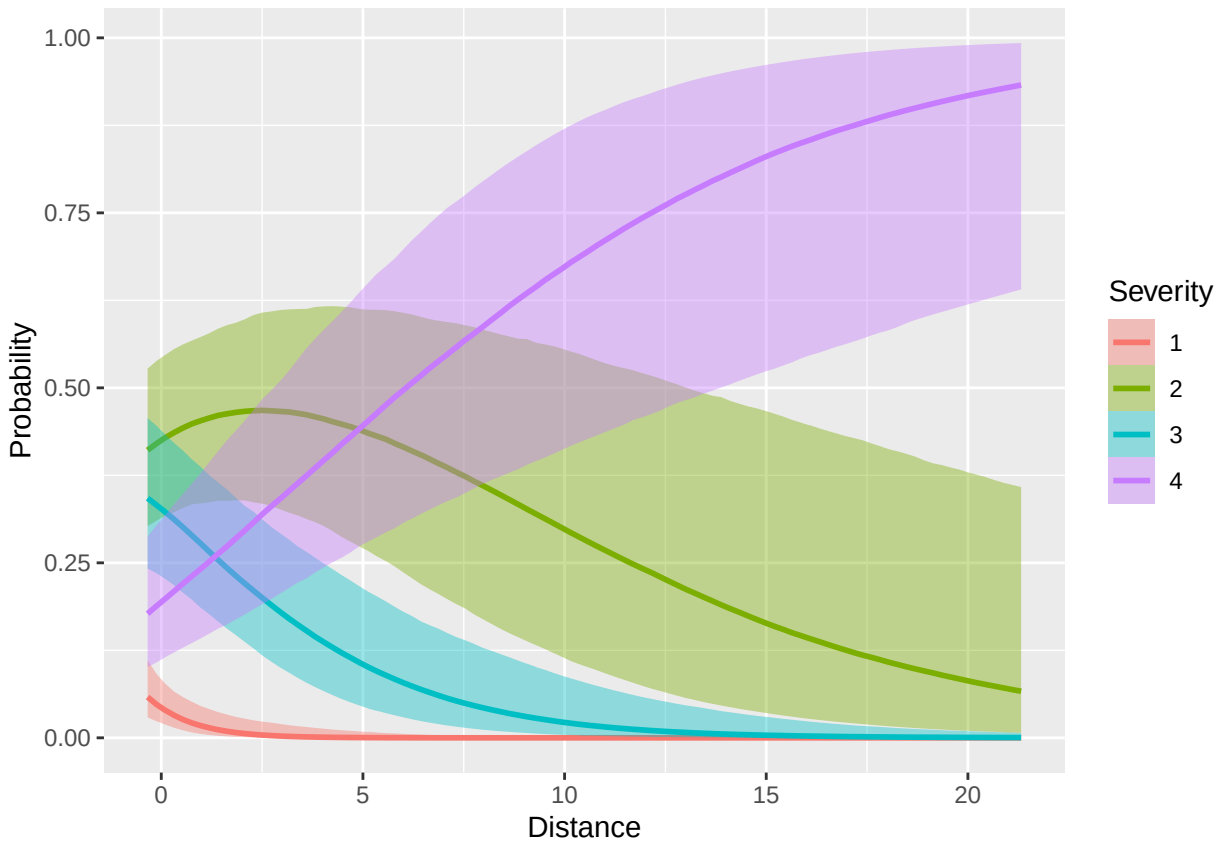

```
plot(conditional_effects(severity_brm, "Weather_Bucket", categorical=TRUE))
```



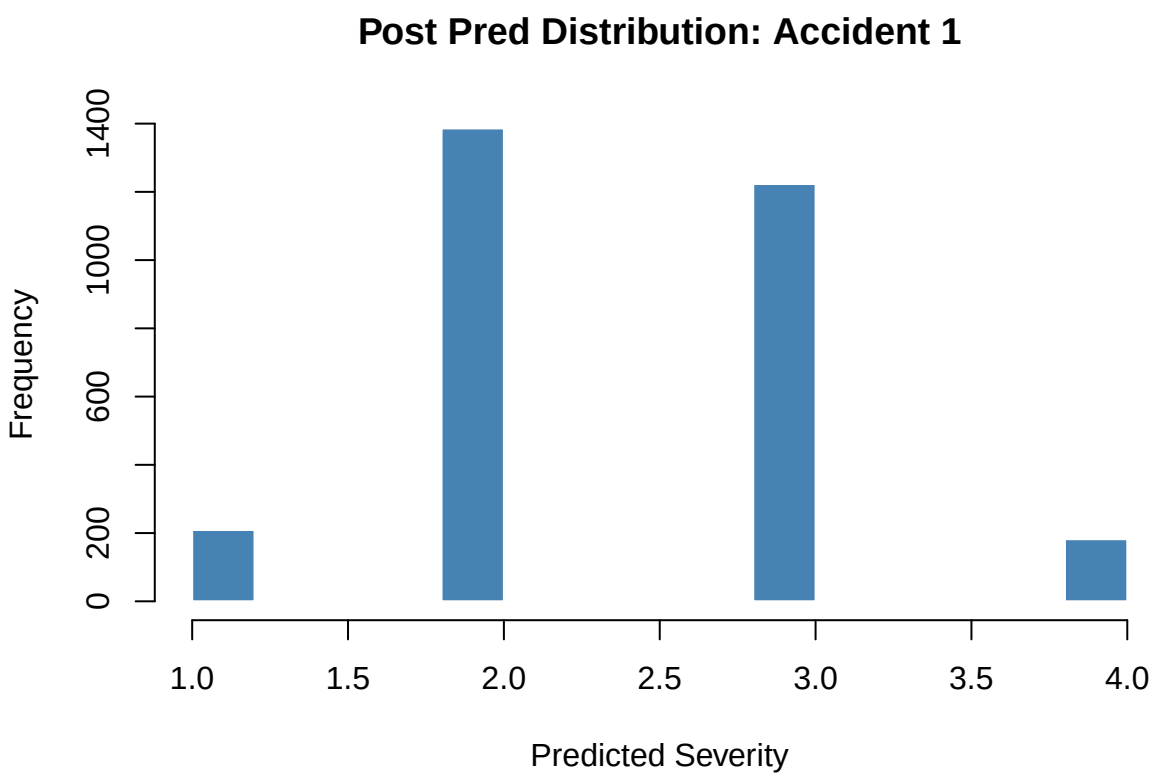

```
plot(conditional_effects(severity_brm, "TimeOfDay", categorical=TRUE))
```




```
plot(conditional_effects(severity_brm, "Distance", categorical=TRUE))
```




```
# posterior predictive distribution for one observation
hist(post_preds[, 1], main="Post Pred Distribution: Accident 1",
     xlab="Predicted Severity", col="steelblue", border="white")
```



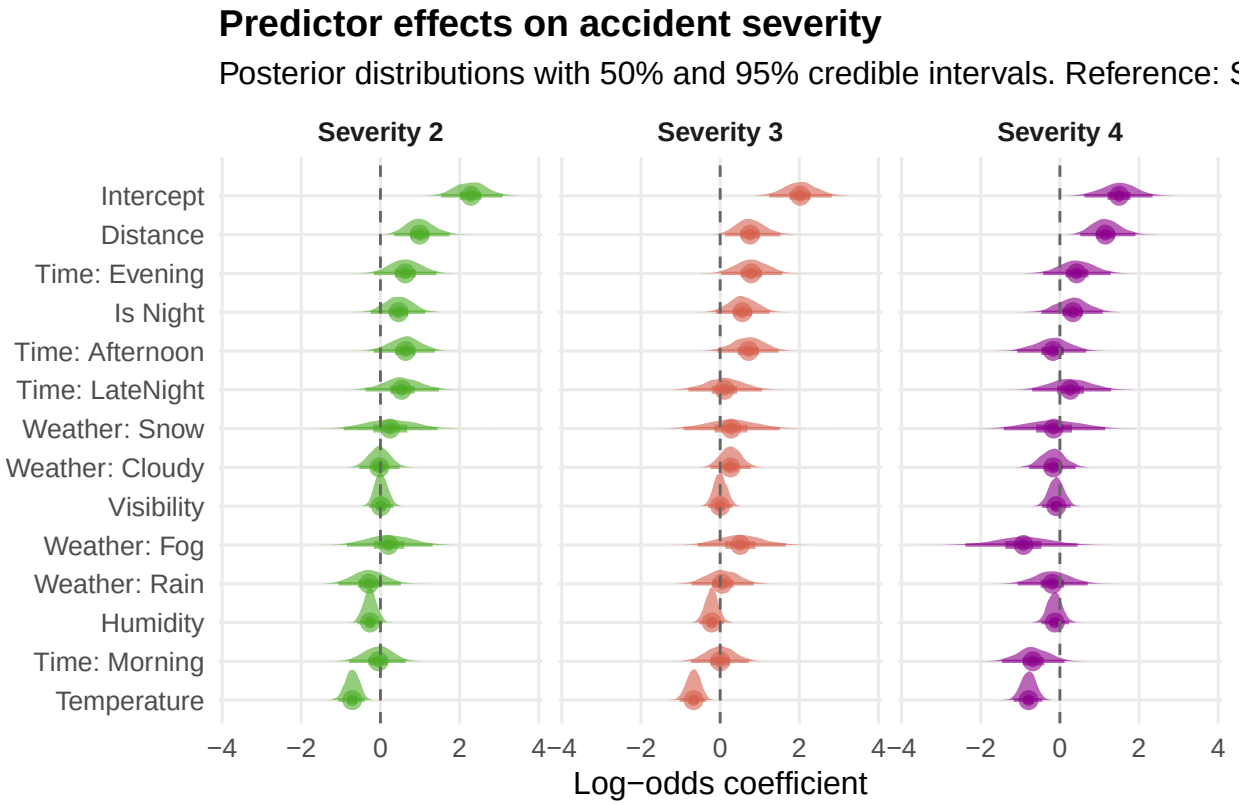
```r
cat("True severity:", as.integer(test$Severity[1]), "\n")
```

```
## True severity: 1
```

### More plotting - to see predictor effects on accident severity

```r
library(tidybayes)
```

```
## Warning: package 'tidybayes' was built under R version 4.3.3
##
## Attaching package: 'tidybayes'
## The following objects are masked from 'package:brms':
##
##     dstudent_t, pstudent_t, qstudent_t, rstudent_t
```

```r
library(ggplot2)
library(dplyr)
library(stringr)

severity_brm %>%
  gather_draws(`b_.*`, regex = TRUE) %>%
  filter(.variable != "b_Intercept") %>%
  mutate(
    # extract severity class from parameter name
    severity_class = case_when(
      str_detect(.variable, "mu2") ~ "Severity 2",
      str_detect(.variable, "mu3") ~ "Severity 3",
      str_detect(.variable, "mu4") ~ "Severity 4"
    ),
    # clean up predictor names for display
    predictor = .variable %>%
      str_remove("b_mu[234]_") %>%
      str_replace_all("Weather_Bucket", "Weather: ") %>%
      str_replace_all("TimeOfDay", "Time: ") %>%
      str_replace_all("IsNightTRUE", "Is Night") %>%
      str_replace_all("Temp", "Temperature") %>%
      str_replace_all("Visibility", "Visibility") %>%
      str_replace_all("Humidity", "Humidity") %>%
      str_replace_all("Distance", "Distance")
  ) %>%
  ggplot(aes(x = .value, y = reorder(predictor, .value), color = severity_class, fill = severity_class)
  stat_halfeye(
    .width = c(0.5, 0.95),
    alpha = 0.6,
    position = position_dodge(width = 0.6)
  ) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray40", linewidth = 0.5) +
  scale_color_manual(values = c("Severity 2" = "#4dac26", "Severity 3" = "#d6604d", "Severity 4" = "#8B
  scale_fill_manual(values  = c("Severity 2" = "#4dac26", "Severity 3" = "#d6604d", "Severity 4" = "#8B
  facet_wrap(~ severity_class, ncol = 3) +
  labs(
    title = "Predictor effects on accident severity",
    subtitle = "Posterior distributions with 50% and 95% credible intervals. Reference: Severity 1.",
```

```
    x = "Log-odds coefficient",
    y = NULL,
    caption = "Effects to the right of zero increase probability of that severity class relative to Seve
  ) +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "none",
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    plot.title = element_text(face = "bold"),
    axis.text.y = element_text(size = 10)
  )
```